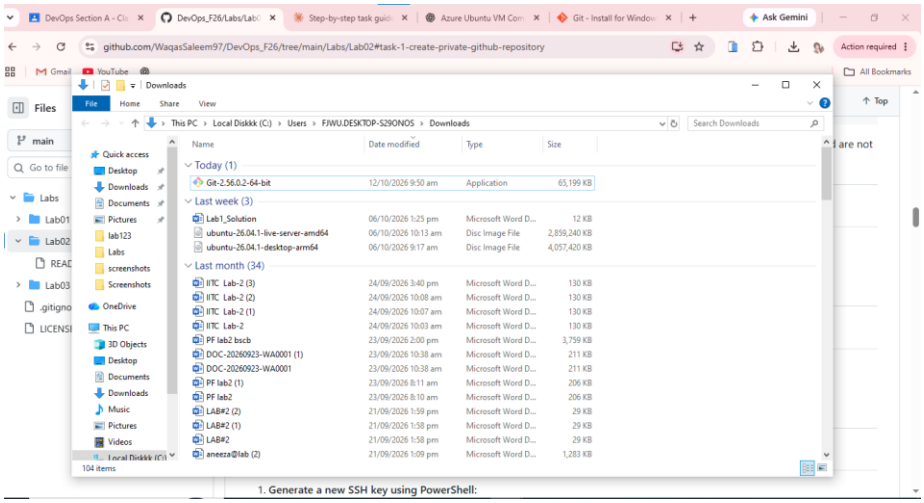
Lab 2: Hands-On Git and Version Control

Name: Sadia GitHub: Sadia612 Course: DevOps

Getting Started: Install Git

Installed Git on Ubuntu and verified the version.

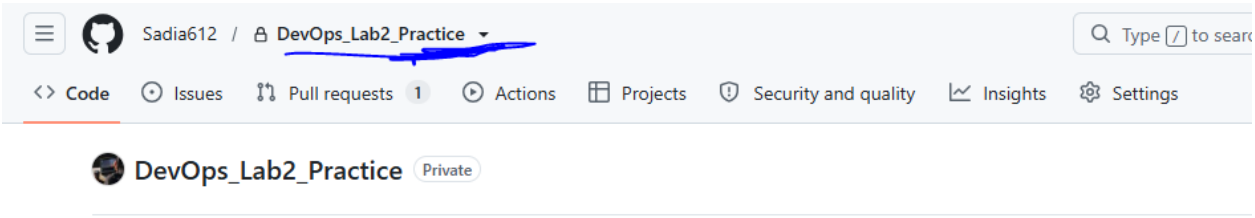
```
sudo apt update sudo
apt install git -y git
--version
```



Task 1: Create Private GitHub Repository

Created a private repository named DevOps_Lab2_Practice on GitHub.

`(GitHub web UI: New repository > Private)`



Task 2: Connect Repository via SSH

Generated an ed25519 key, added the public key to GitHub and cloned over SSH.

```
ssh-keygen -t ed25519 -C "dehr44058@gmail.com"
cat ~/.ssh/id_ed25519.pub ssh -T
git@github.com
git clone git@github.com:Sadia612/DevOps_Lab2_Practice.git
```

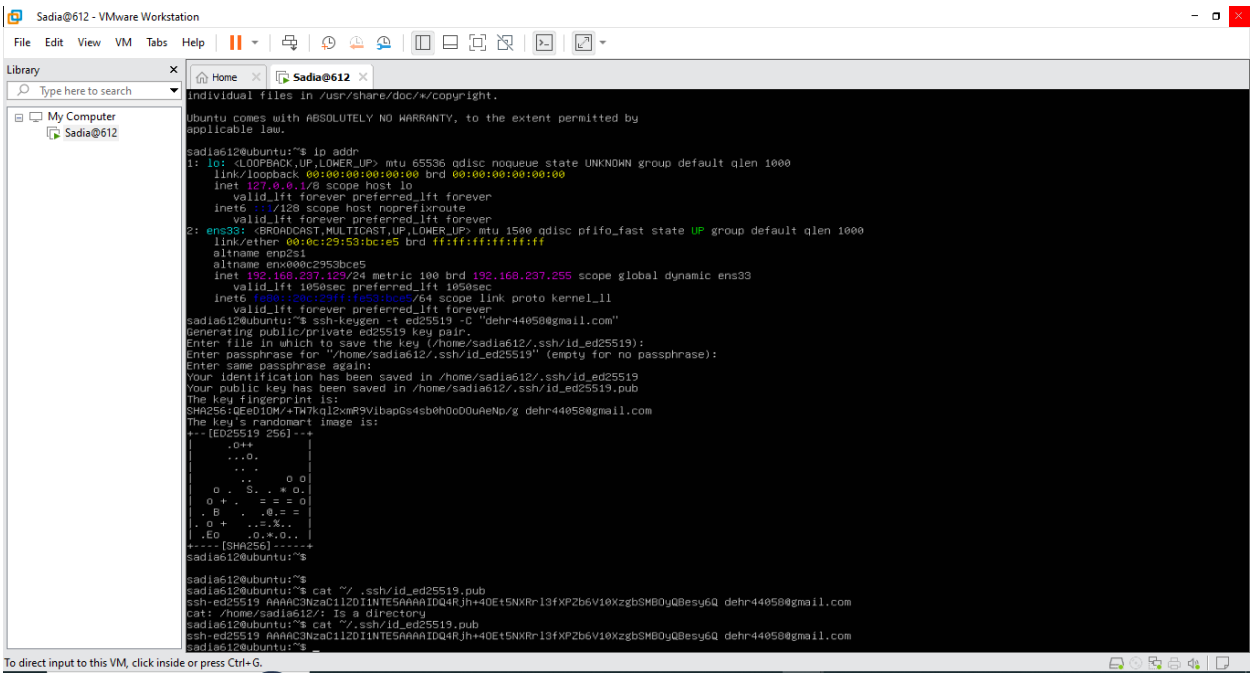


Figure: SSH key generation (ssh_keygen.png)

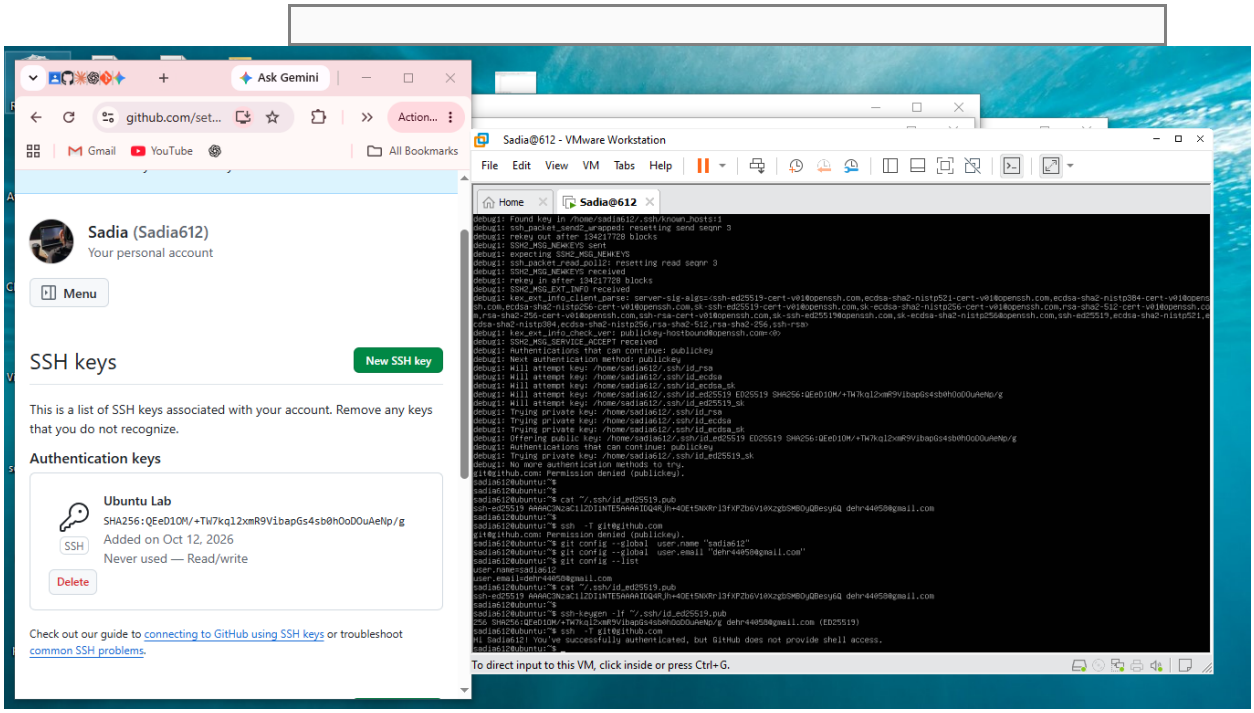


Figure: Public key added to GitHub (github_sshkey.png)

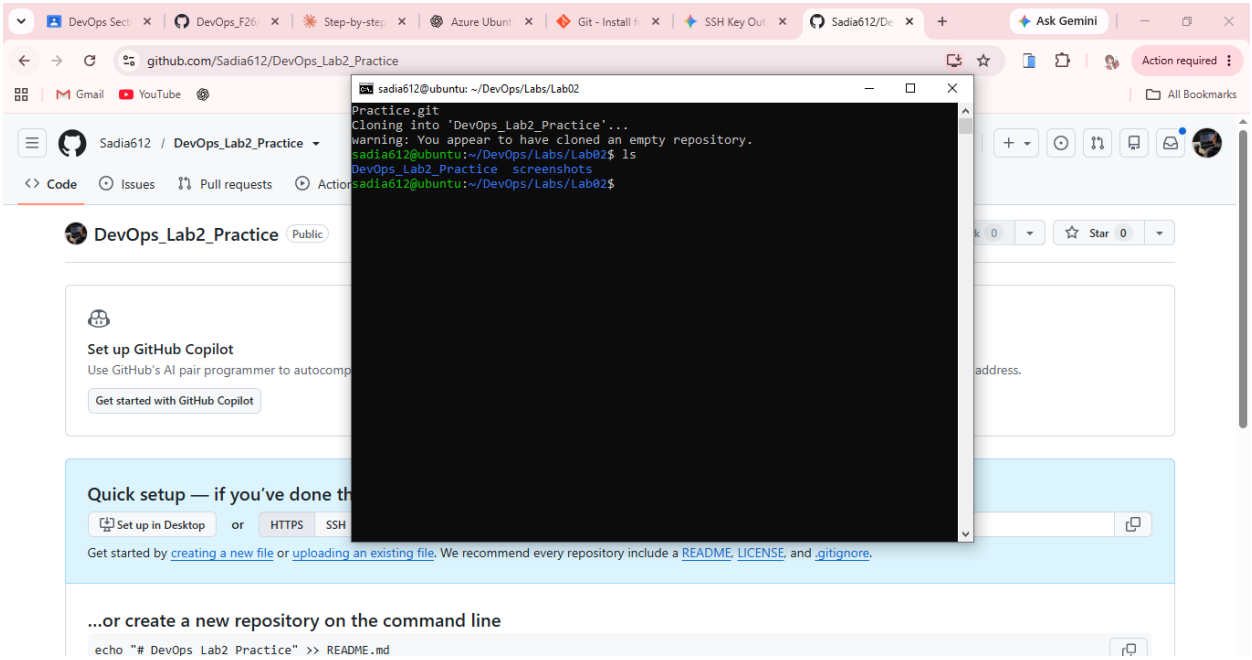


Figure: Clone via SSH (ssh_clone.png)

Task 3: Configure Git Username and Email

Set the global identity used for commits.

```
git config --global user.name "Sadia" git config --
global user.email "dehr44058@gmail.com" git config -
-list
```

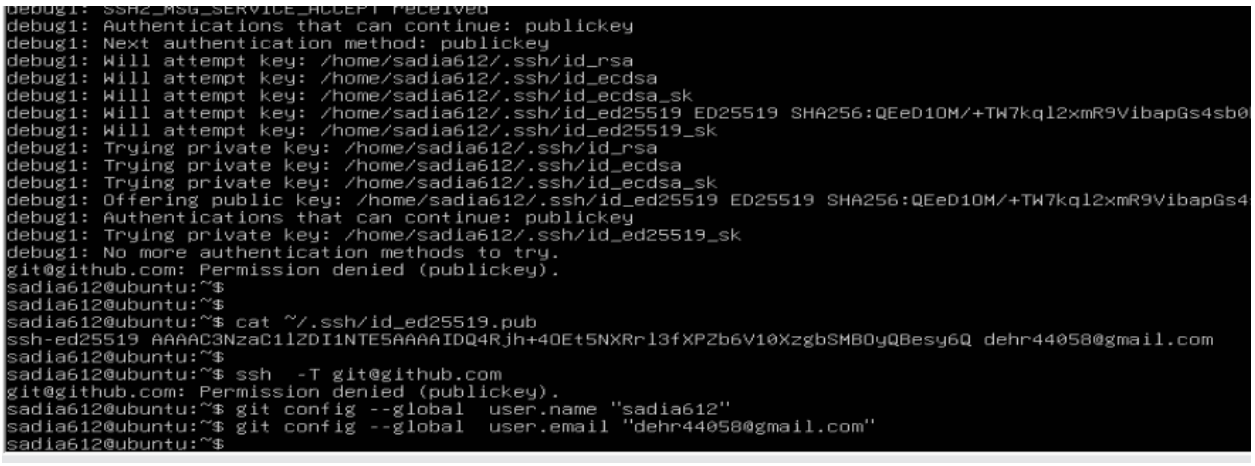


Figure: Git identity (git_identity.png)

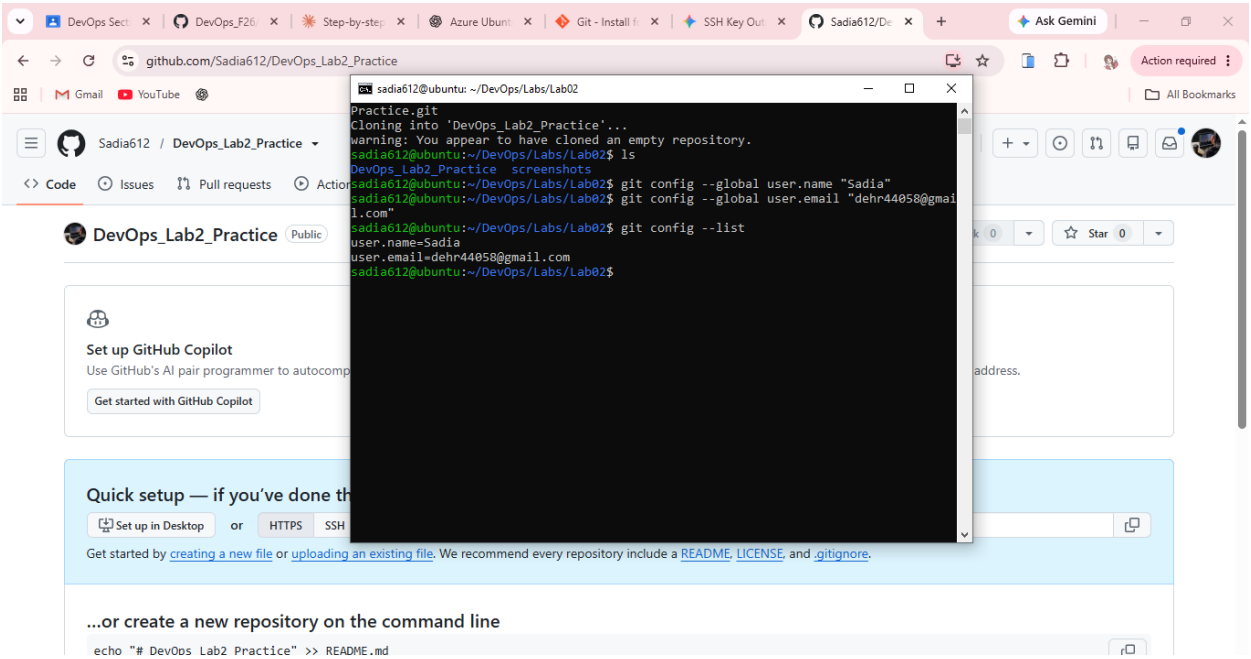


Figure: git config --list (git_config_list.png)

Task 4: Explore the .git Folder

The .git folder stores history, config, refs and objects of the repository.

```
cd DevOps_Lab2_Practice
ls -a ls -a .git
```

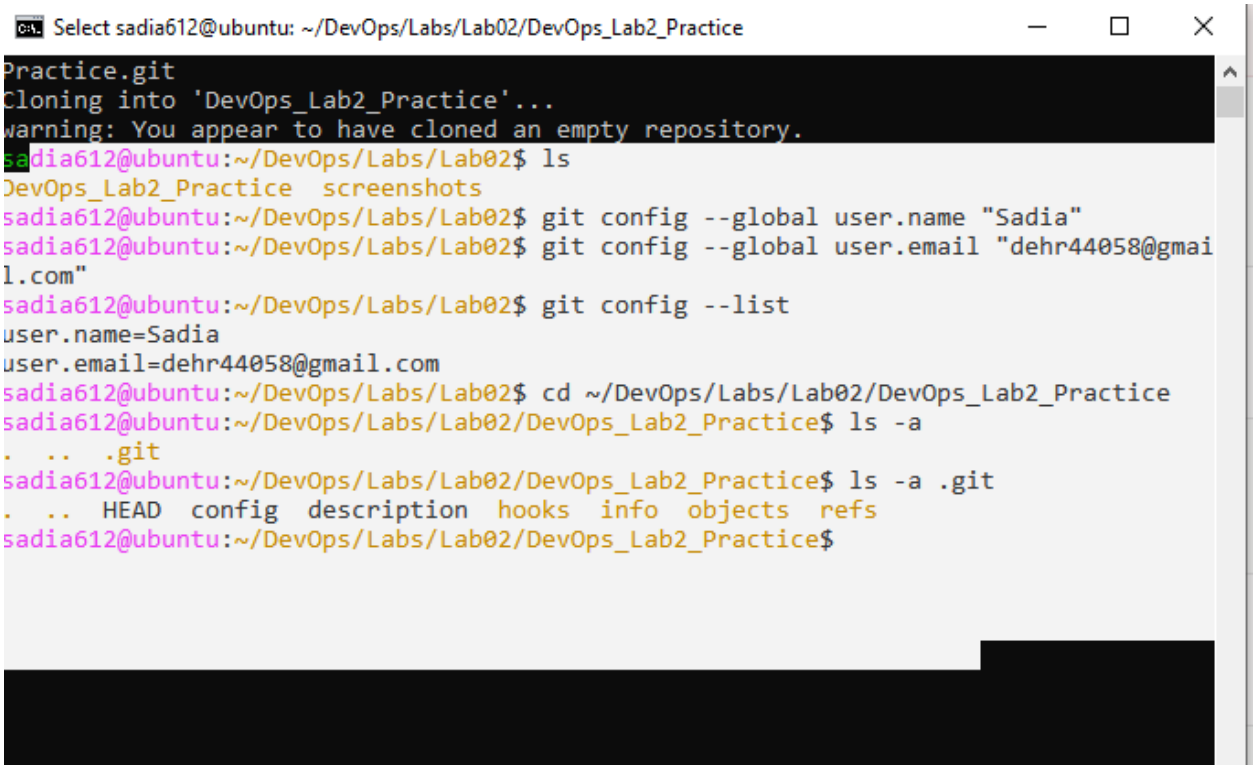


Figure: .git folder contents (git_folder.png)

Task 5: Local Repository Management

After rm -rf .git, git status fails with 'fatal: not a git repository' because all repository data (history, config, refs) lives inside .git; without it the folder is just a normal directory.

```
rm -rf .git git status git init echo "# Lab02 Git Practice" >>
README.md git add README.md git commit -m "Initial commit" git remote
add origin git@github.com:Sadia612/DevOps_Lab2_Practice.git git branch
-M main git push -u origin main
```

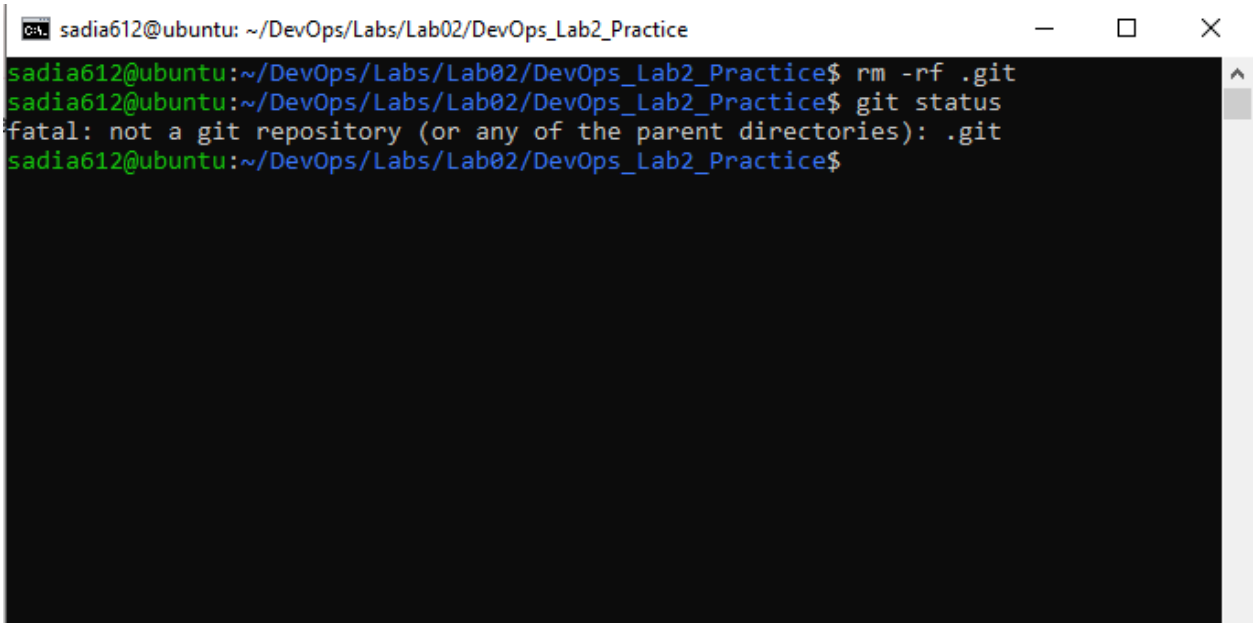


Figure: .git deleted, git status error (delete_git.png)

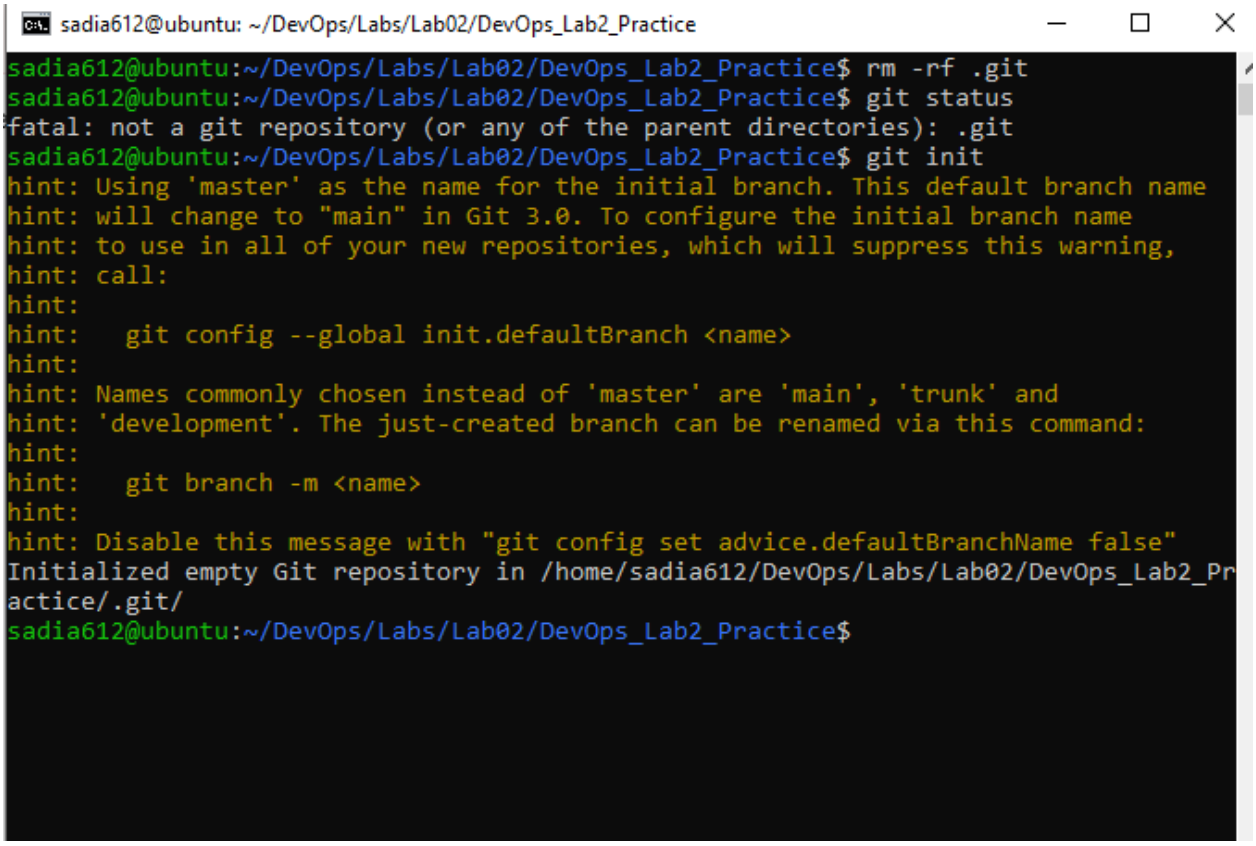


Figure: git init (git_init.png)

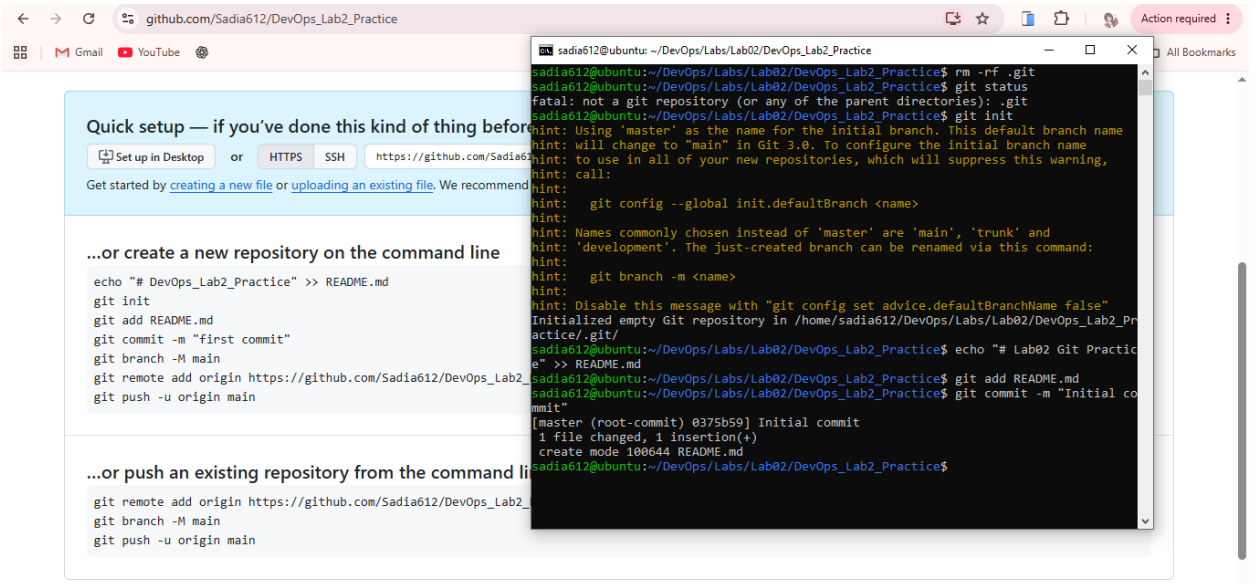


Figure: First commit (first_commit.png)

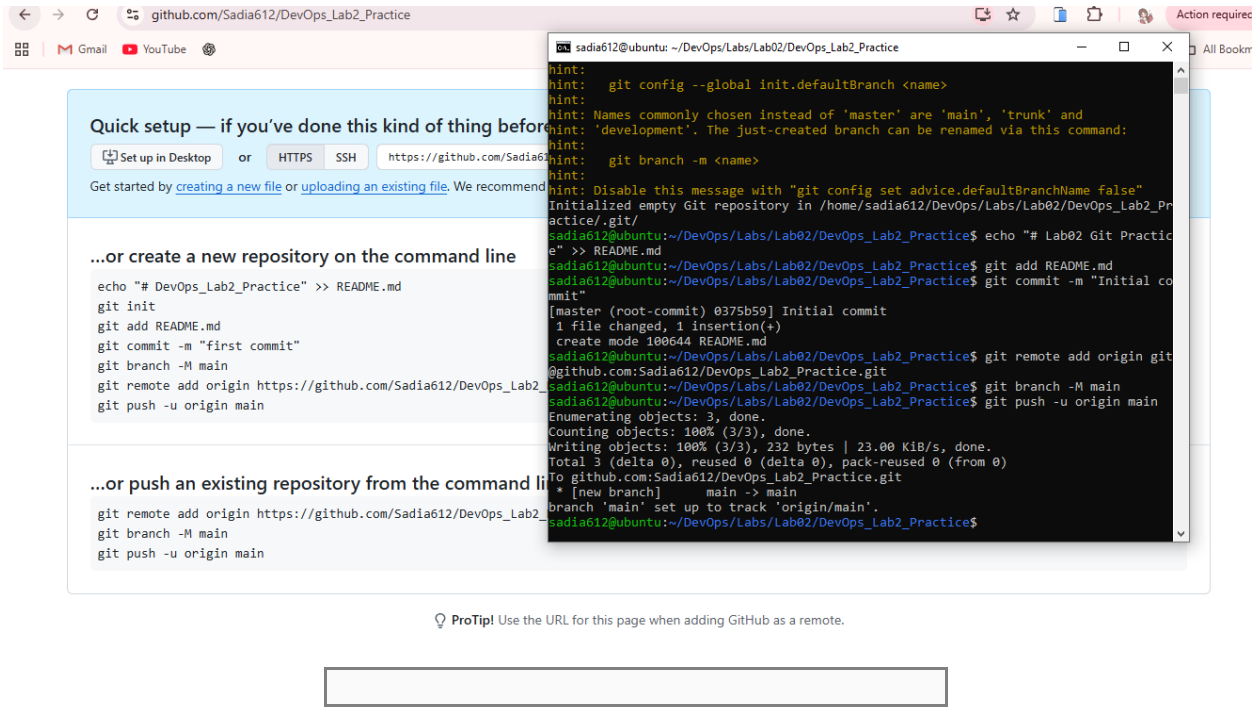


Figure: First push (first_push.png)

Task 6: File Status & Staging

Created notes.txt, checked status, staged and committed.

```
echo "Adding some test notes" >> notes.txt
git status git add notes.txt git commit -m
"Add notes.txt"
```

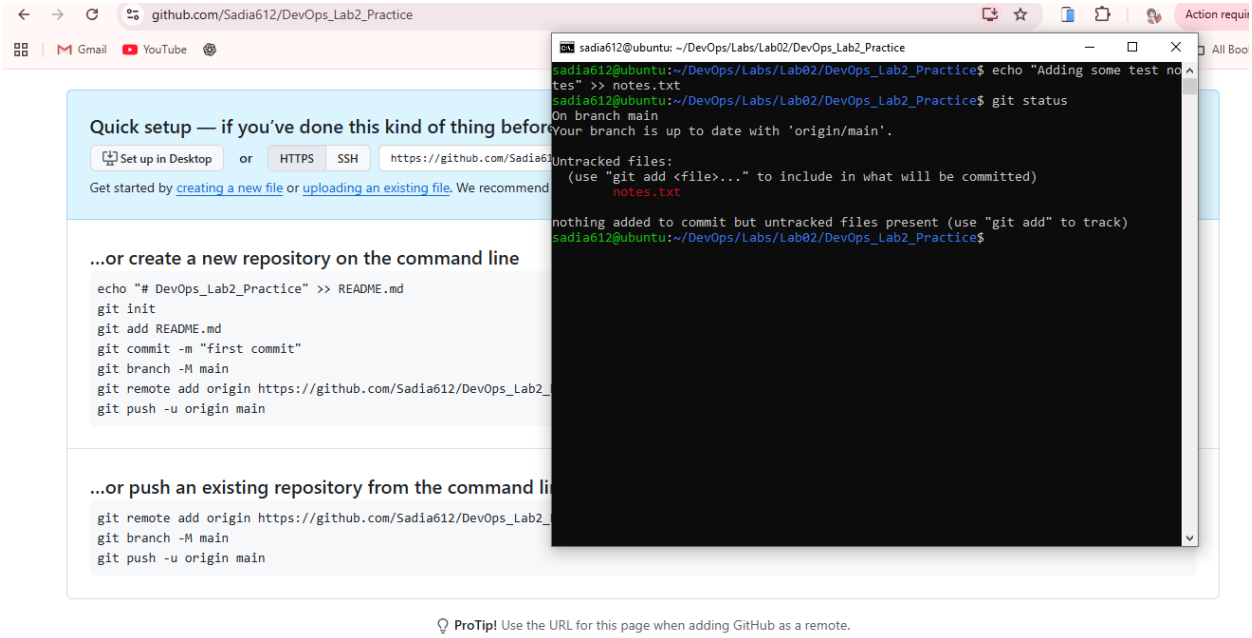


Figure: Untracked notes.txt (status1.png)

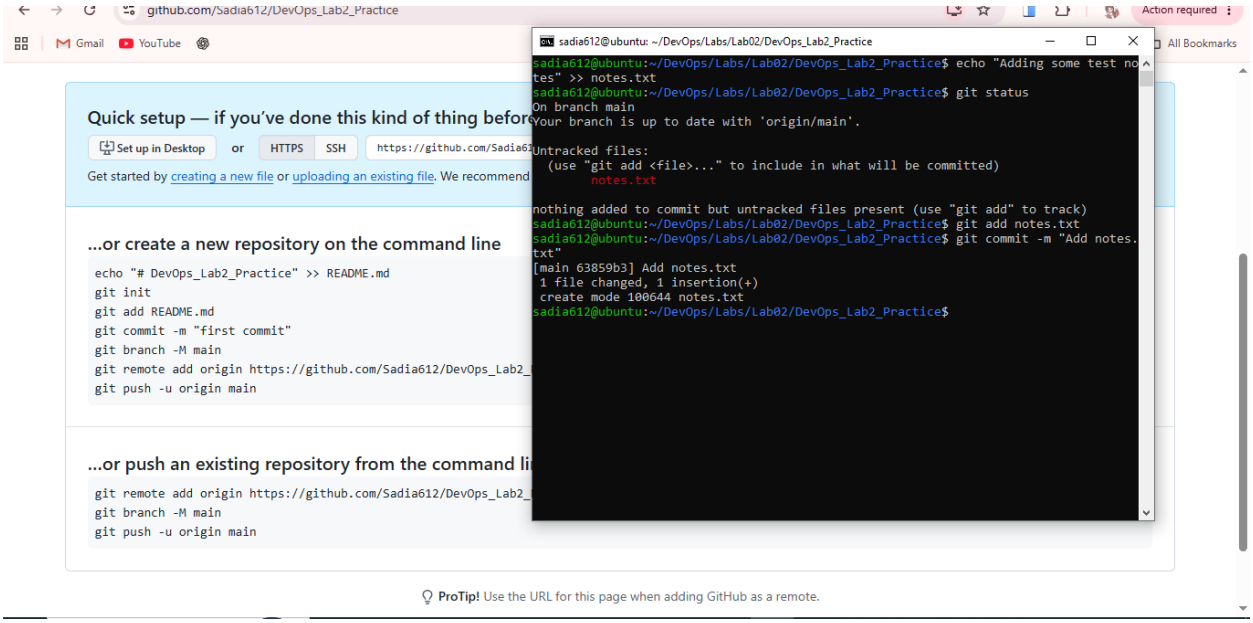


Figure: Commit of notes.txt (commit_notes.png)

Task 7: Branch Creation Using GitHub GUI

Created bugfix/user-auth-error on GitHub and pulled it locally.

```
git pull origin bugfix/user-auth-error
git branch -a
```

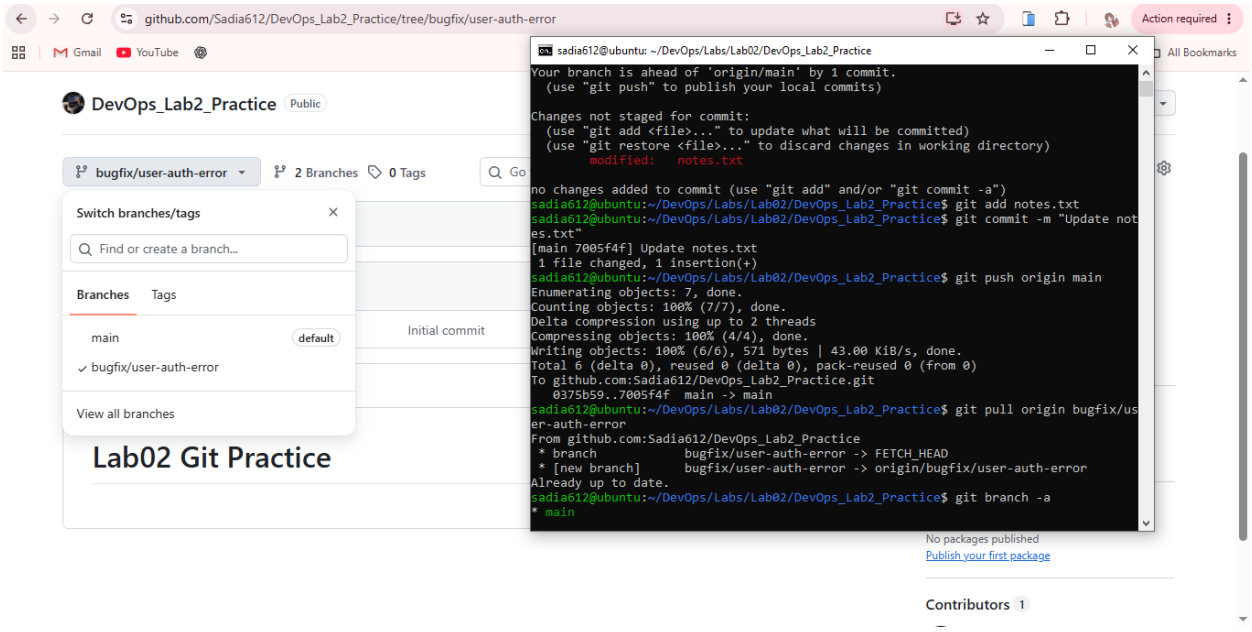


Figure: Branch created in GitHub GUI (bugfix_branch_gui.png)

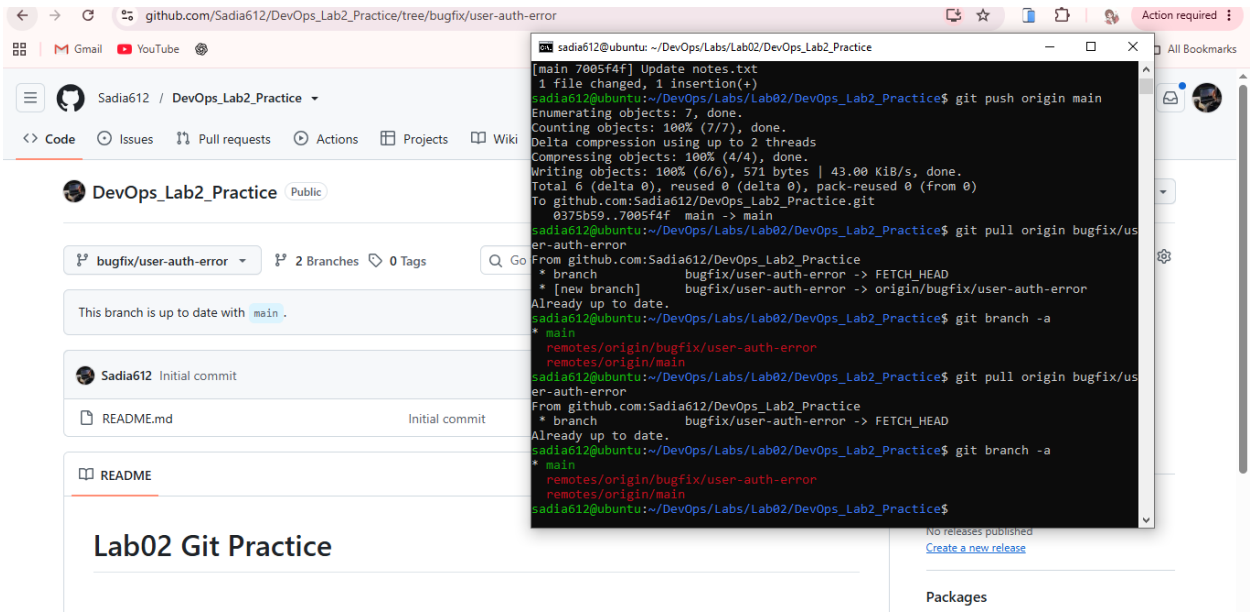


Figure: Branch list after pull (bugfix_branch_local.png)

Task 8: Branch Creation and Push Using Git Bash

Created feature/db-connection locally and pushed it.

```
git checkout -b feature/db-connection
git push origin feature/db-connection
```

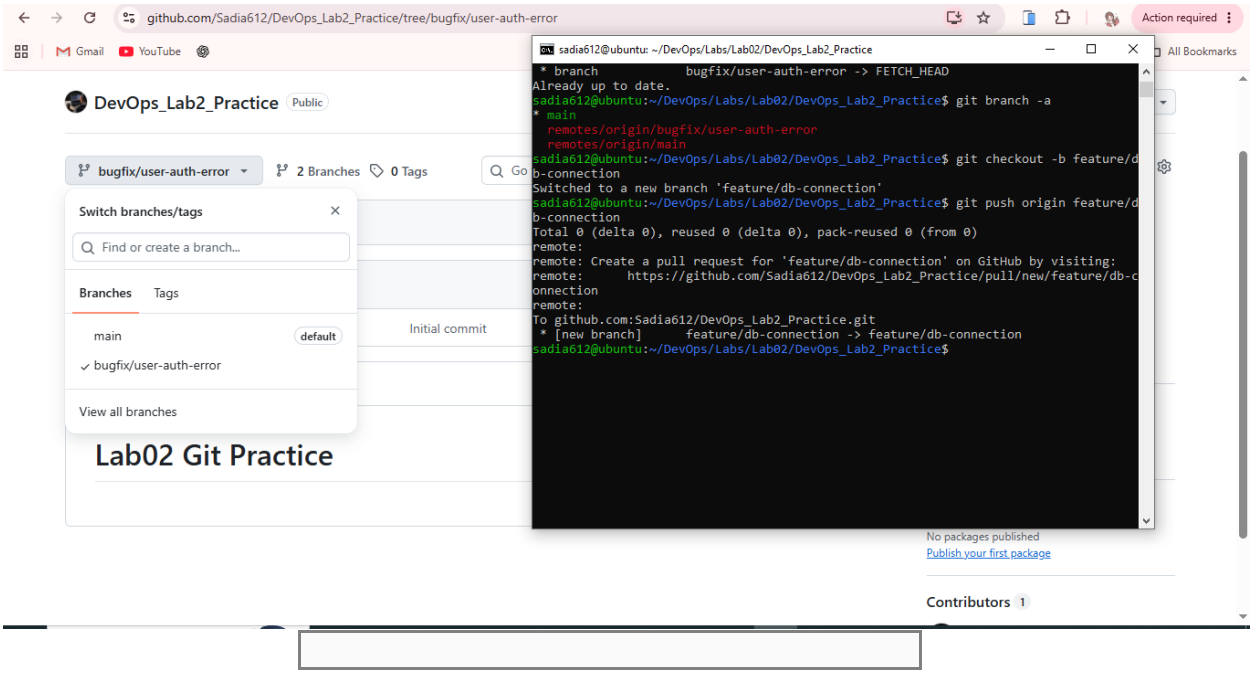


Figure: feature/db-connection created and pushed (feature_db_branch.png)

Task 9: Branching & Merging

Created feature-1, added a function to main.py, merged into main and pushed.

```
git checkout -b feature-1 git add main.py
git commit -m "Add new function to main.py"
git checkout main git merge feature-1 git
push origin main git push origin feature-1
```

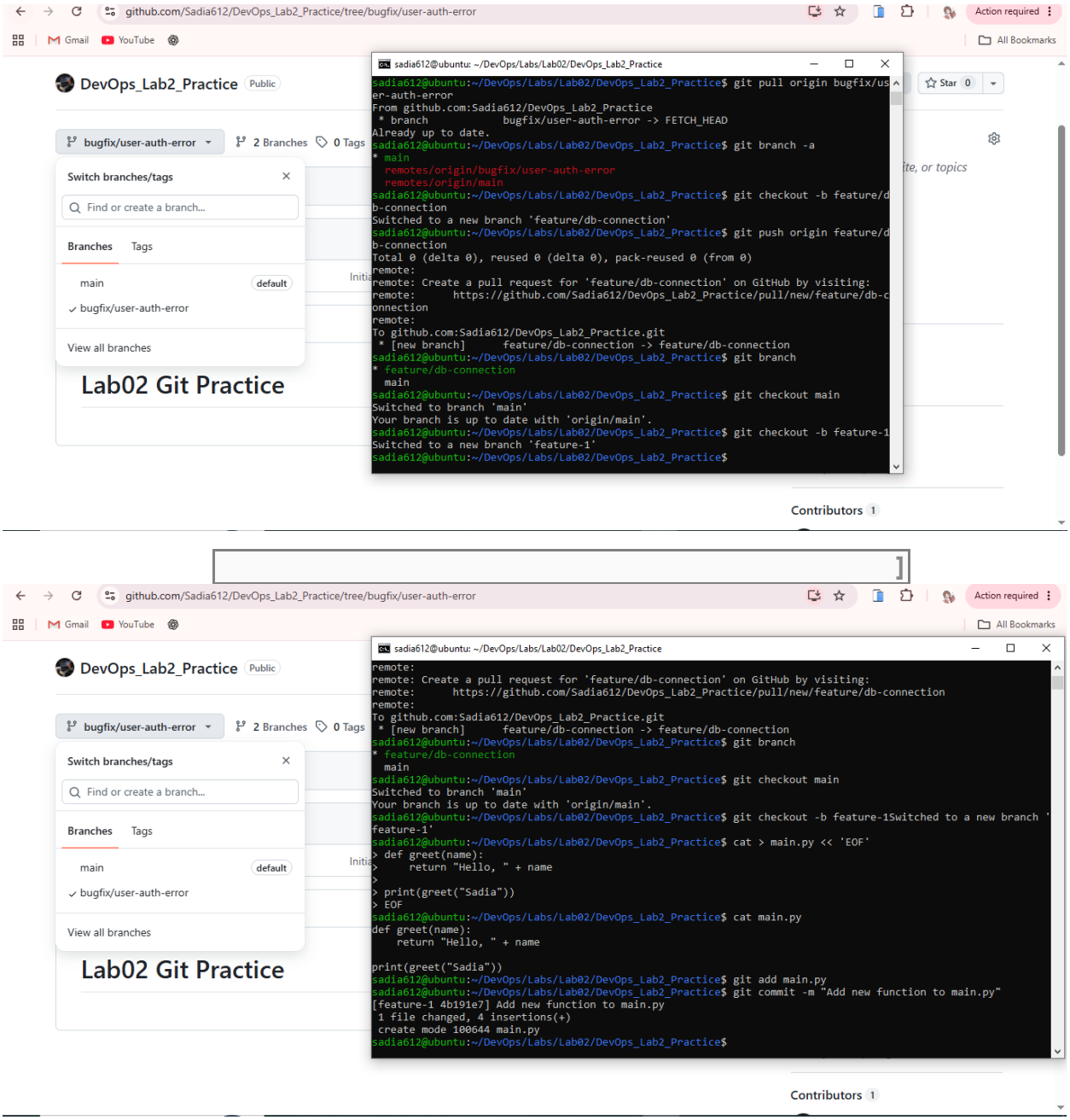


Figure: Commit on feature-1 (feature_commit.png)

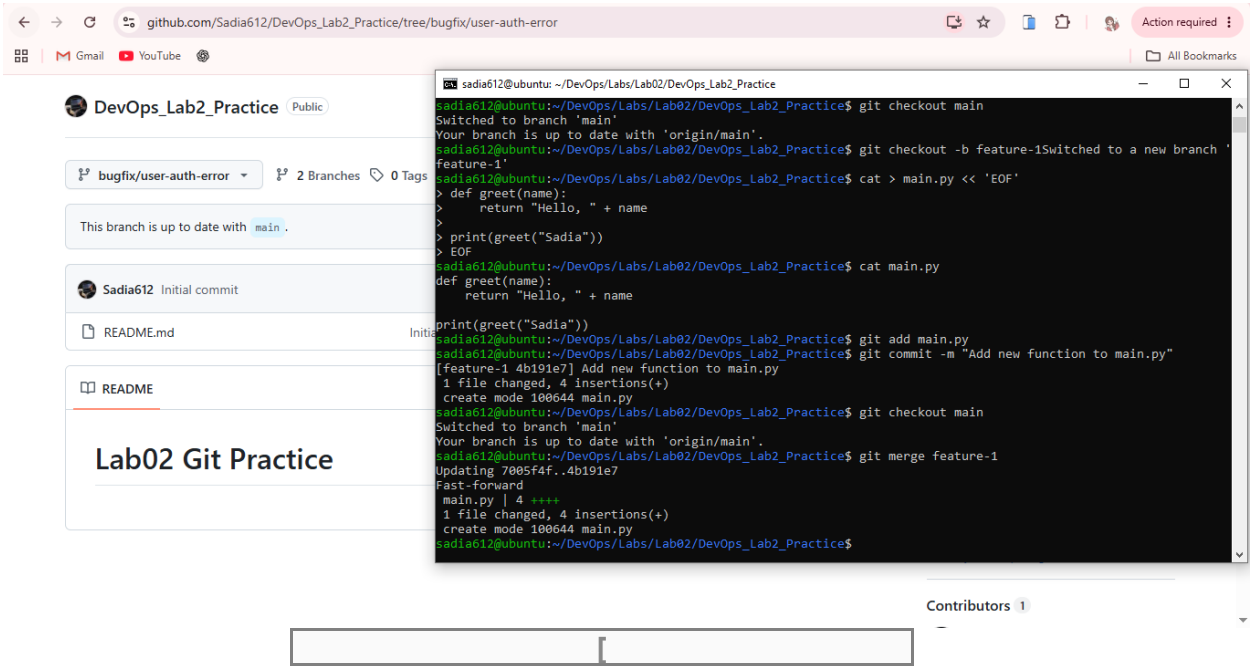


Figure: Merge into main (merge.png)

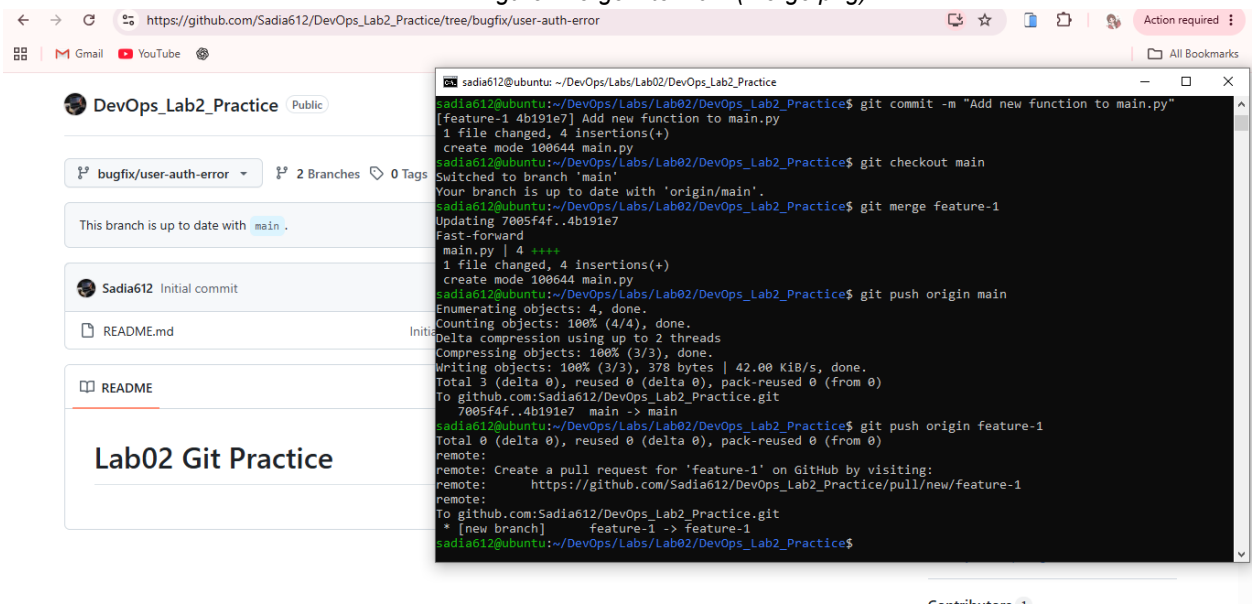


Figure: Push of main and feature-1 (push_branches.png)

Task 10: Pull Request and Branch Review (GitHub GUI)

Opened a PR from feature/db-connection to main, reviewed and merged it, then deleted the branch.

`(GitHub web UI)`

[SCREENSHOT PLACEHOLDER]

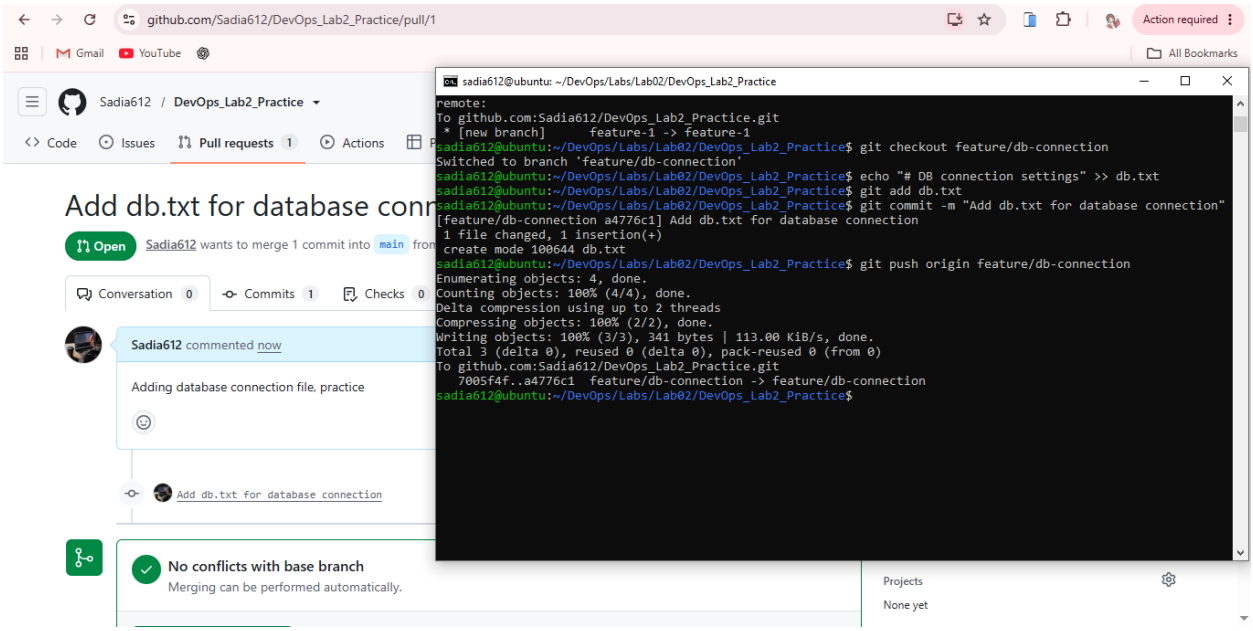
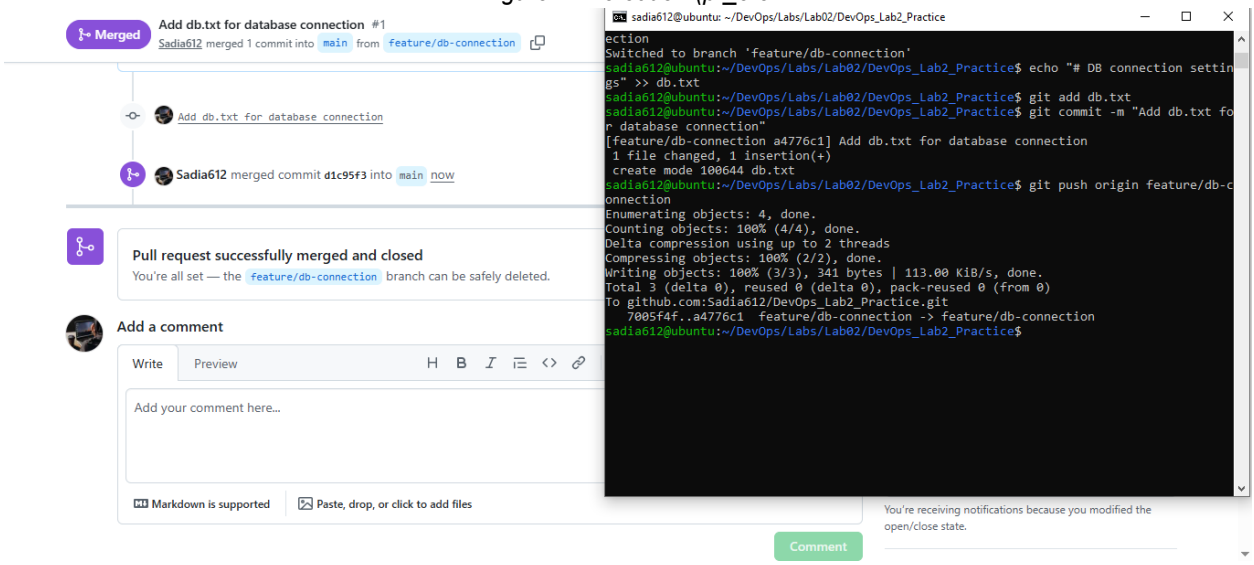


Figure: PR creation (pr_cre



ation.png)

[SCREENSHOT PLACEHOLDER]

Figure: PR review and merge (pr_merge.png)

[SCREENSHOT PLACEHOLDER]

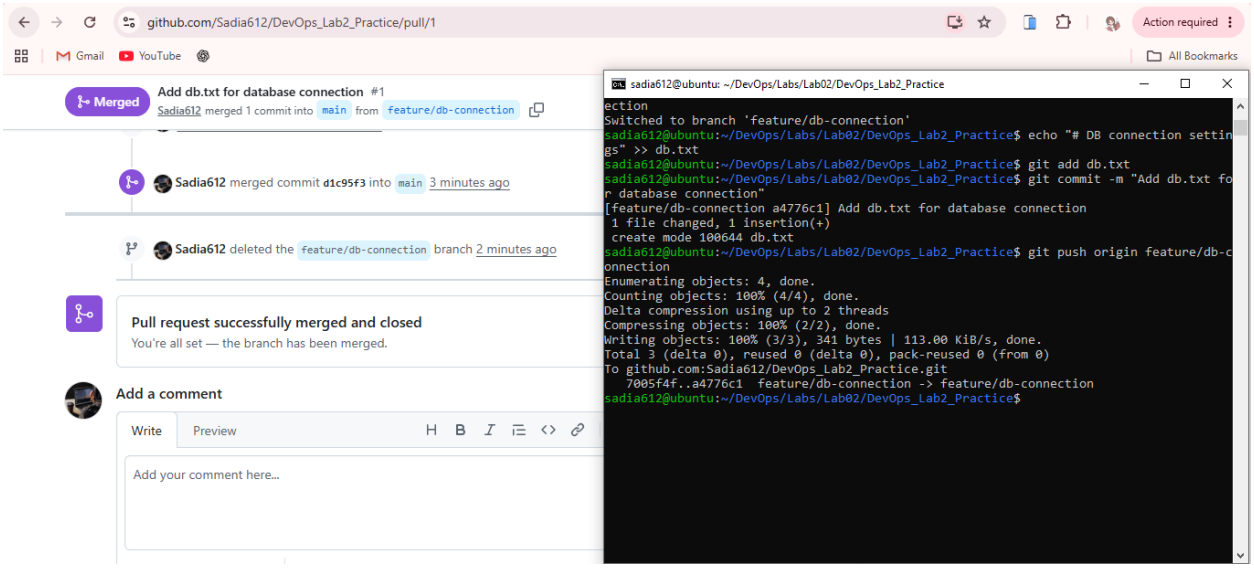


Figure: Branch deletion (branch_delete.png)

Task 11: Detailed Branch Strategy (Develop/Staging)

Flow: feature/* and bugfix/* -> develop -> staging -> main.

```
git checkout -b develop git checkout
-b staging git merge feature/login
(on develop) git merge develop (on
staging) git merge staging (on main)
git log --oneline --graph --all
```

```
sadia612@ubuntu: ~/DevOps/Labs/Lab02/DevOps_Lab2_Practice
Fast-forward
 db.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 db.txt
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout -b develop
Switched to a new branch 'develop'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git push origin develop
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/Sadia612/DevOps_Lab2_Practice/pull/new/develop
remote:
To github.com:Sadia612/DevOps_Lab2_Practice.git
 * [new branch]      develop -> develop
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout -b staging
Switched to a new branch 'staging'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git push origin staging
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'staging' on GitHub by visiting:
remote:      https://github.com/Sadia612/DevOps_Lab2_Practice/pull/new/staging
remote:
To github.com:Sadia612/DevOps_Lab2_Practice.git
 * [new branch]      staging -> staging
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout -b feature/login
Switched to a new branch 'feature/login'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout -b bugfix/typo
Switched to a new branch 'bugfix/typo'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git branch
  bugfix/typo
  develop
  feature-1
  feature/db-connection
  feature/login
* main
  staging
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$
```

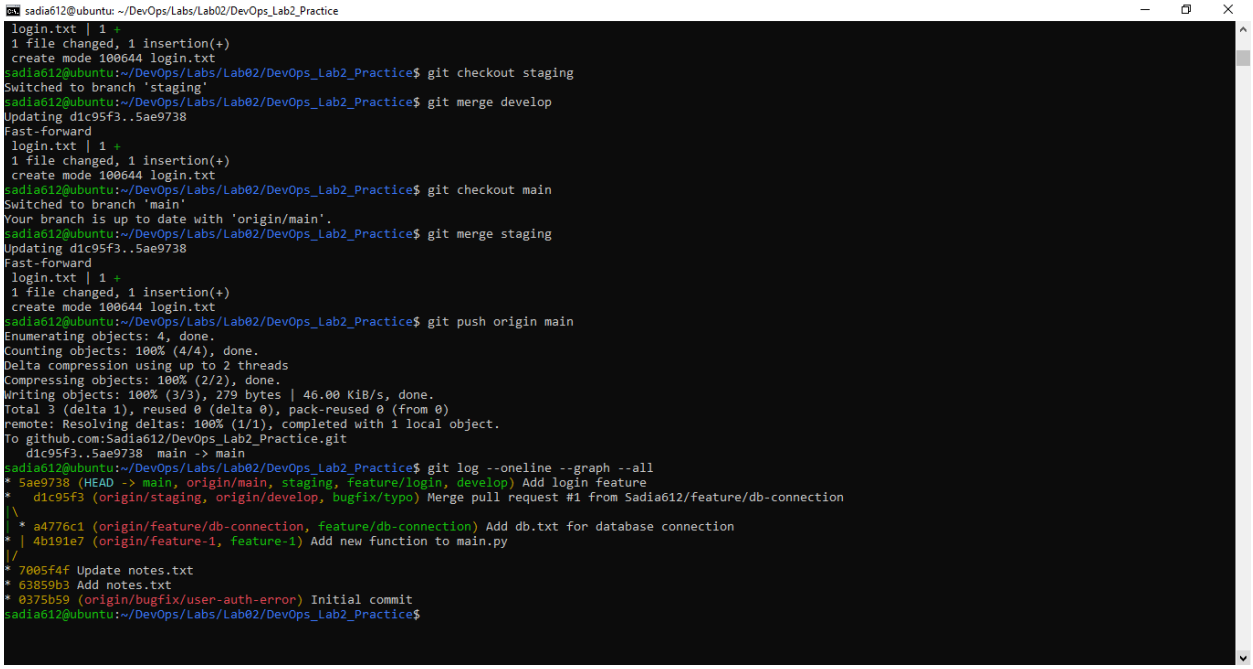
[SCREENSHOT PLACEHOLDER]

Figure: Branch list (branch_strategy.png)

```
sadia612@ubuntu: ~/DevOps/Labs/Lab02/DevOps_Lab2_Practice
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout -b feature/login
Switched to a new branch 'feature/login'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout -b bugfix/typo
Switched to a new branch 'bugfix/typo'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git branch
  bugfix/typo
  develop
  feature-1
  feature/db-connection
  feature/login
* main
  staging
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout feature/login
Switched to branch 'feature/login'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ echo "login feature" >> login.txt
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git add login.txt
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git commit -m "Add login feature"
[feature/login 5ae9738] Add login feature
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git merge feature/login
Updating d1c95f3..5ae9738
Fast-forward
 login.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git checkout staging
Switched to branch 'staging'
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$ git merge develop
Updating d1c95f3..5ae9738
Fast-forward
 login.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
sadia612@ubuntu:~/DevOps/Labs/Lab02/DevOps_Lab2_Practice$
```

[SCREENSHOT PLACEHOLDER]

Figure: Merges into develop and staging (branch_merges.png)



[SCREENSHOT PLACEHOLDER]

Figure: Final merge into main (final_merge.png)

Task 12: Code Review Workflow

PR #2 (feature/profile -> main) with title and description, reviewer assigned, changes requested, fixed with a new commit, approved and merged. A separate practice PR was closed to show rejection.

```
git checkout -b feature/profile git commit -m
"Add profile feature" git push origin
feature/profile git commit -m "Add docstring to
profile function" git push origin feature/profile
```

[SCREENSHOT PLACEHOLDER]

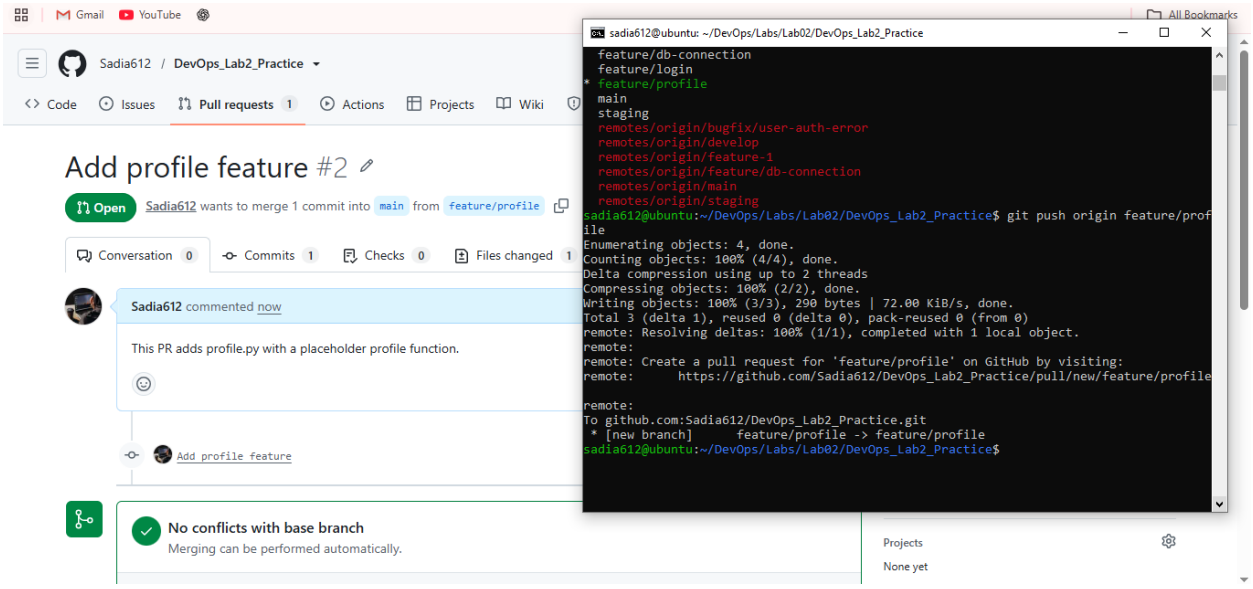
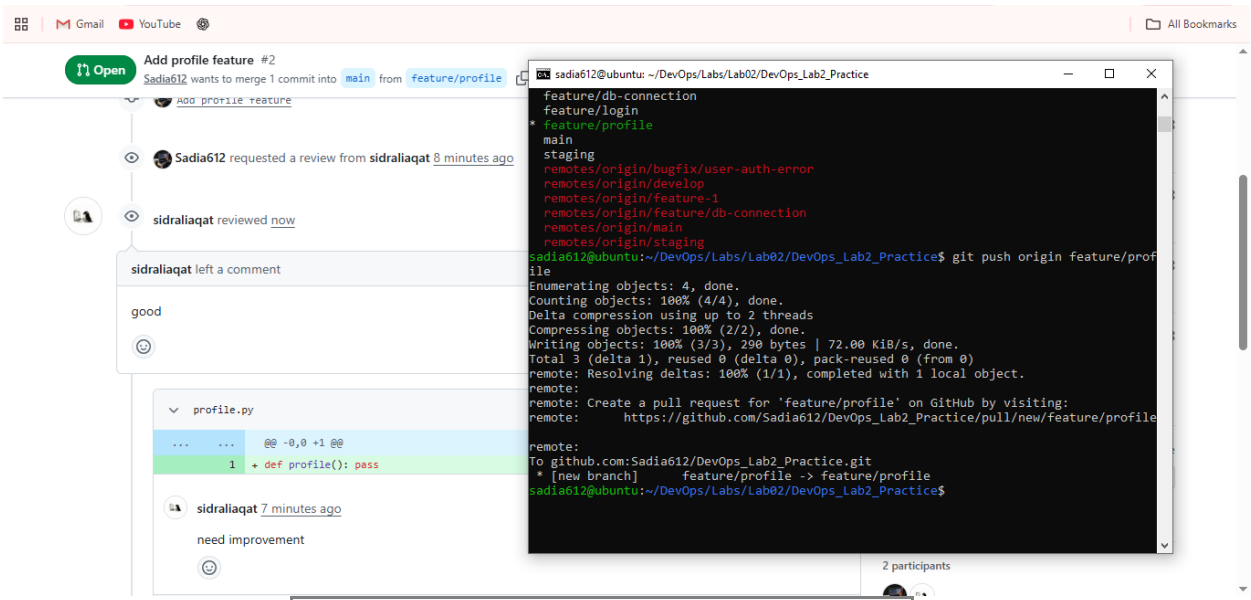
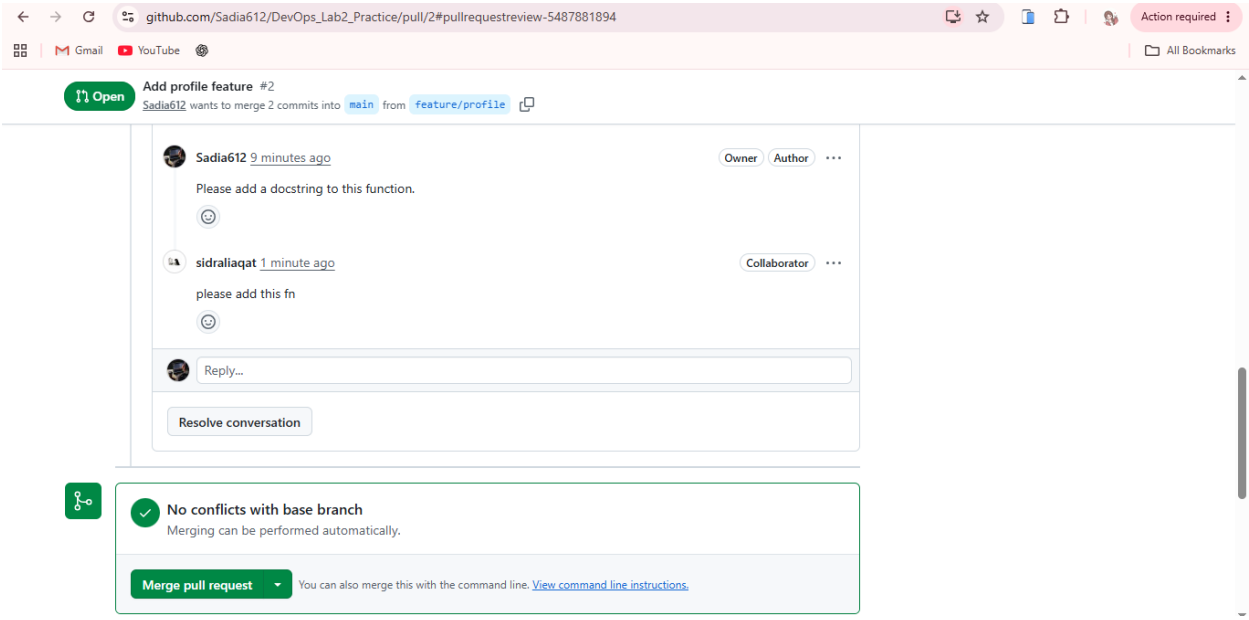


Figure: PR title and description (pr_create_details.png)



[SCREENSHOT PLACEHOLDER]

Figure: Reviewer assigned (pr_assigned_reviewer.png)



[SCREENSHOT PLACEHOLDER] *Figure: Reviewer requested changes (pr_request_changes.png)*

[SCREENSHOT PLACEHOLDER]

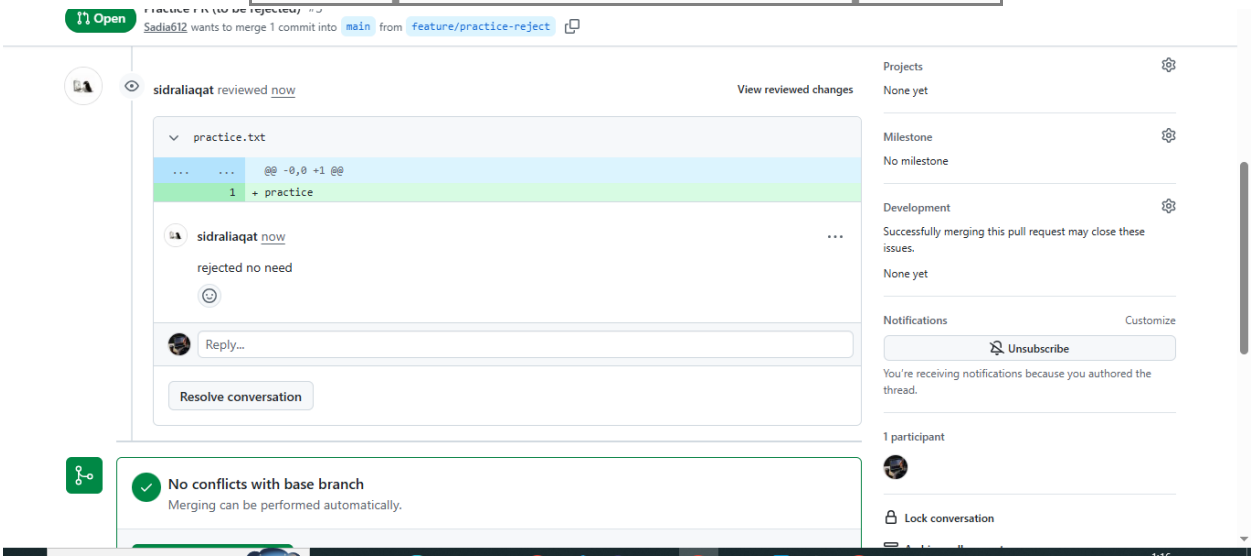


Figure: Practice PR closed (pr_rejected.png)

[SCREENSHOT PLACEHOLDER]

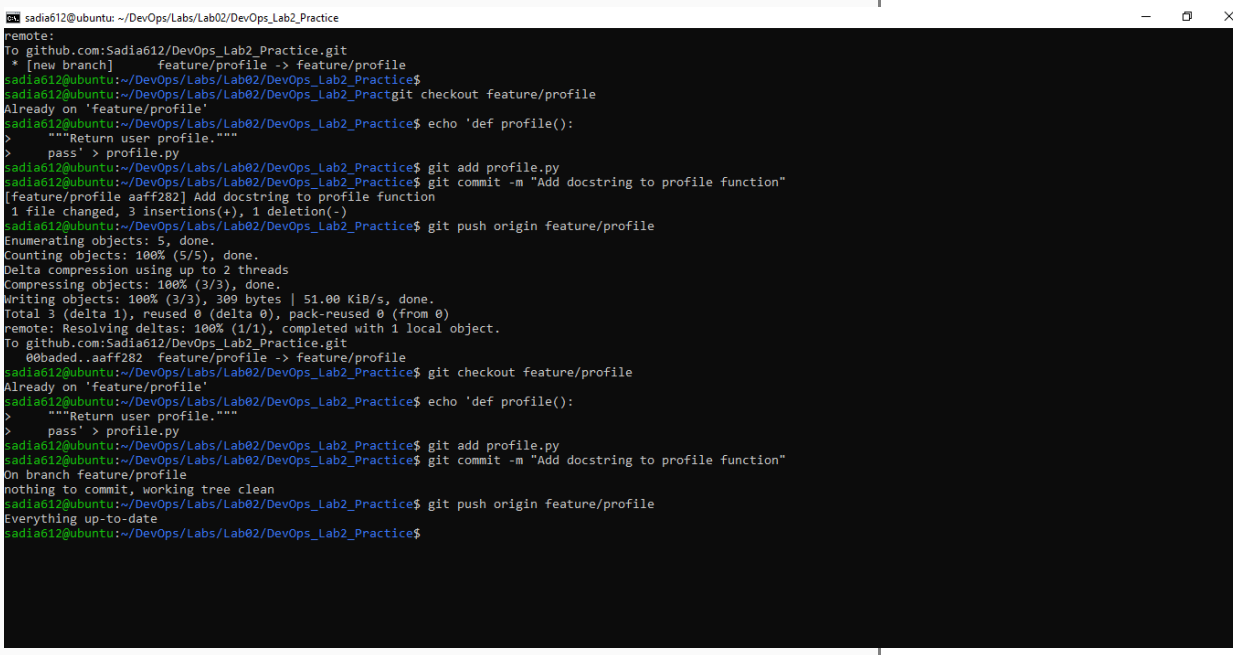
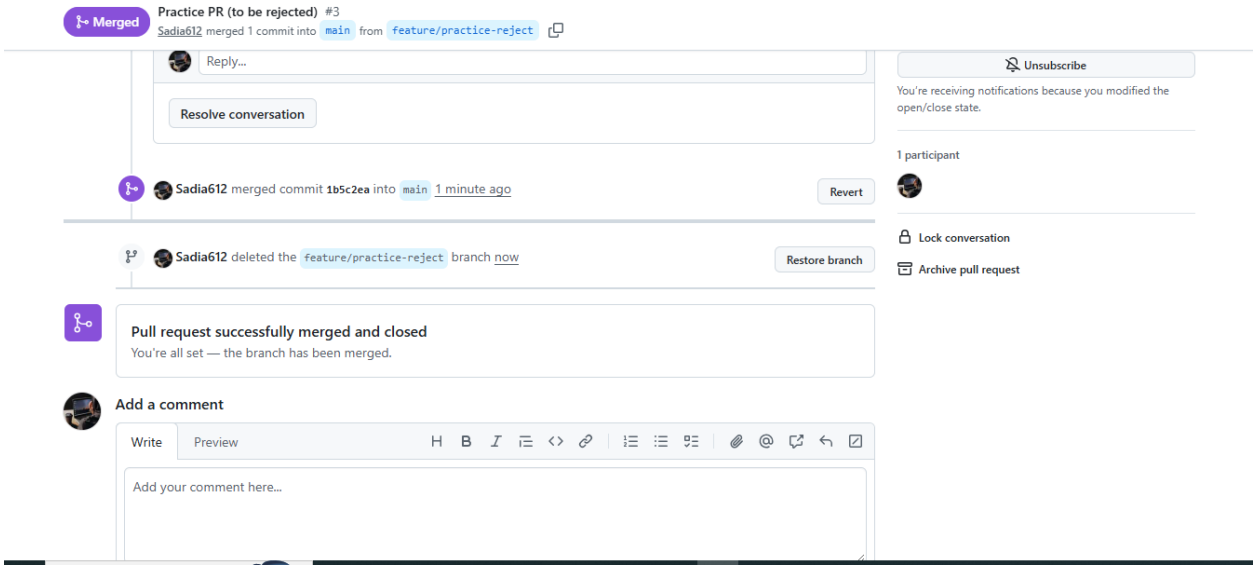


Figure: PR updated with new commit (pr_updated_with_commits.png)



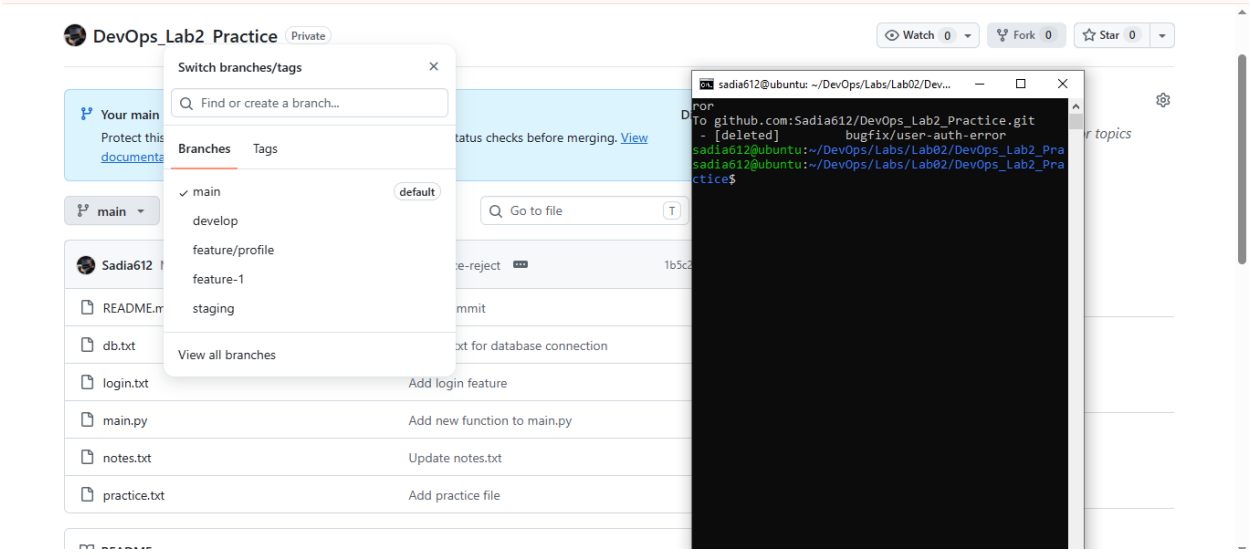
[SCREENSHOT PLACEHOLDER]

Figure: Merged branch deleted (pr_branch_deleted.png)

Task 13: Branch Cleanup Best Practices

Deleted merged branches from GitHub UI and from the terminal, then cleaned local refs.

```
git push origin --delete <branch-name>
git checkout main git pull origin main
git fetch --prune git branch -d
<branch-name> git branch
```



[SCREENSHOT PLACEHOLDER]

Figure: Remote branch deleted via GitHub UI (remote_branch_deleted.png)

[SCREENSHOT PLACEHOLDER]

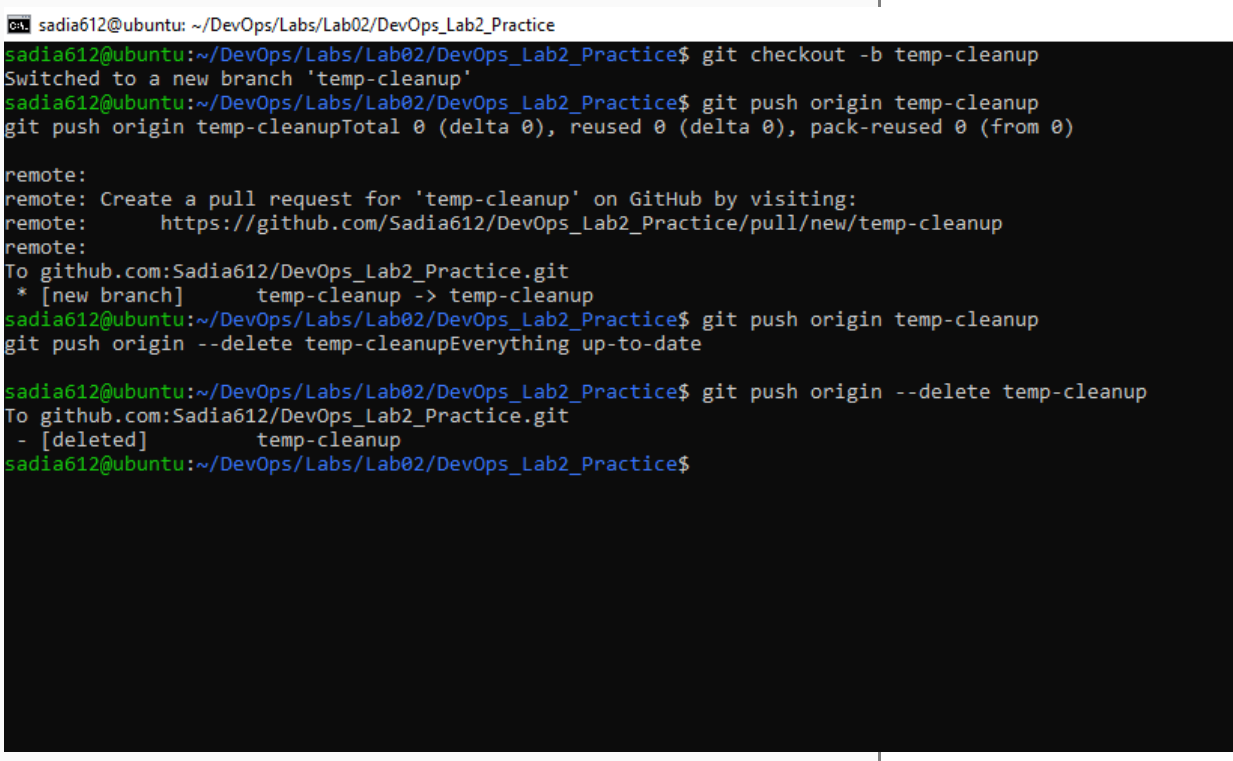


Figure: Remote branch deleted via terminal (remote_branch_delete_cmd.png)